

# Micturition

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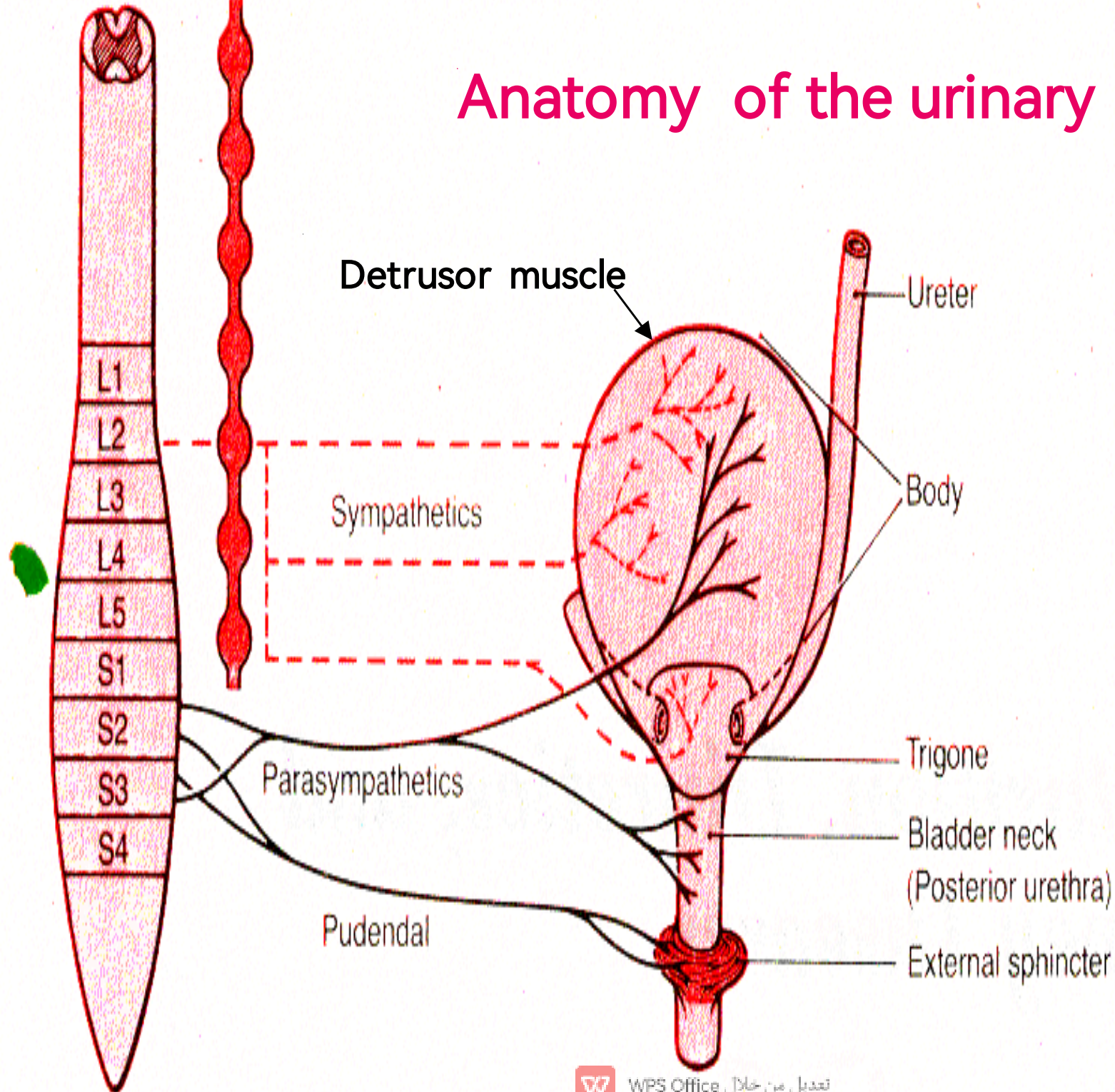


تعديل من خلال WPS Office

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- Micturition is a process by which urine is voided from the urinary bladder.
  - It is a reflex process.
  - The functional anatomy and nerve supply of urinary bladder are essential for the process of micturition.



# Anatomy of the urinary bladder



**Figure 31-1.** Urinary bladder innervation.

**TABLE 57.1: Functions of nerves supplying urinary bladder and sphincters**

| Nerve                 | On detrusor muscle | On internal sphincter | On external sphincter | Function                         |
|-----------------------|--------------------|-----------------------|-----------------------|----------------------------------|
| Sympathetic nerve     | Relaxation         | Constriction          | Not supplied          | Filling of urinary bladder       |
| Parasympathetic nerve | Contraction        | Relaxation            | Not supplied          | Emptying of urinary bladder      |
| Somatic nerve         | Not supplied       | Not supplied          | Constriction          | Voluntary control of micturition |



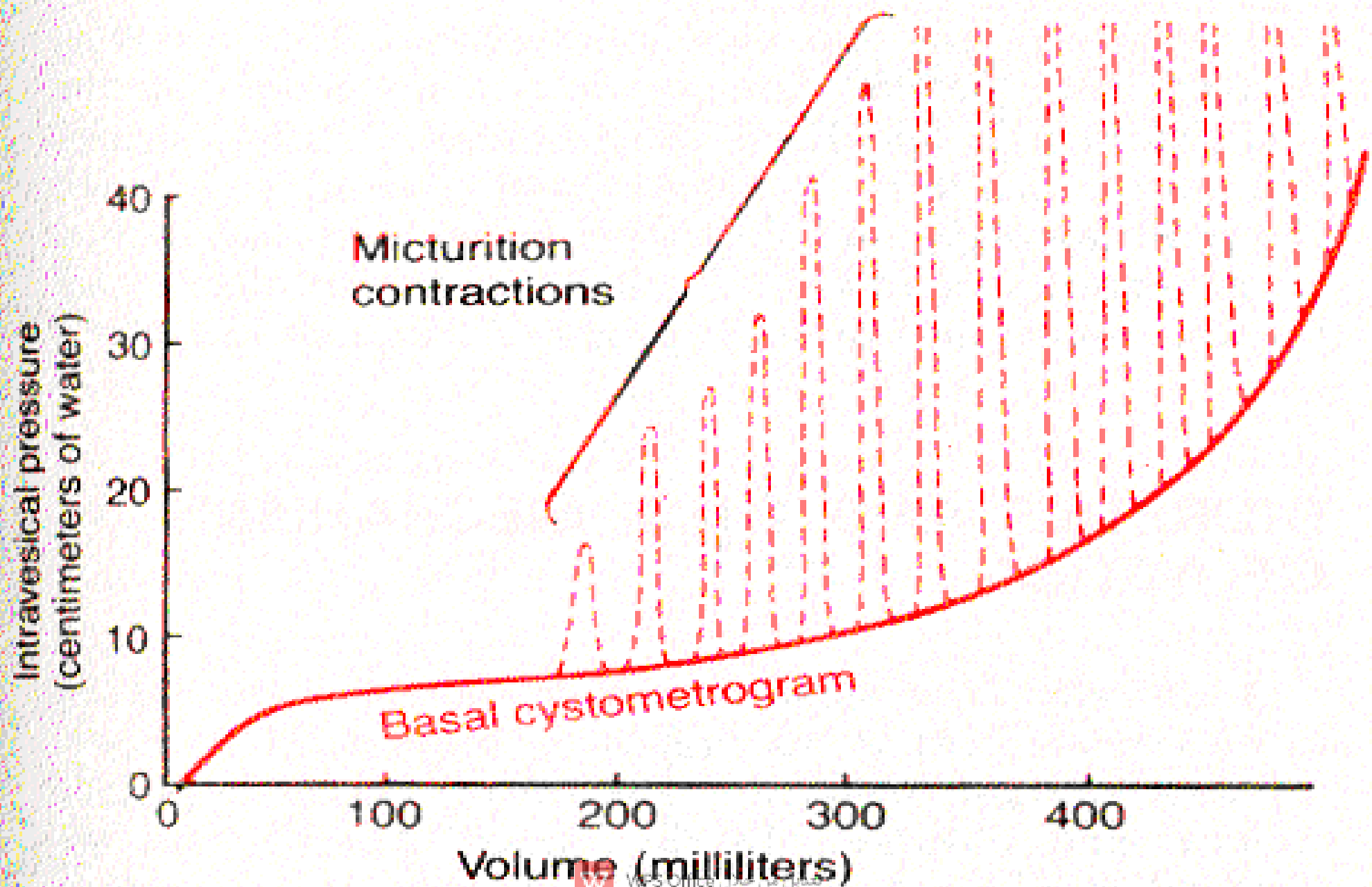


# Filling of the urinary bladder

- peristaltic contractions facilitate the passage of urine from renal pelvis through ureters to urinary bladder.



# cystometrogram



Normal cystometrogram showing also contractions

# Micturition Reflex

- receptors: stretch receptors within the detrusor muscle stimulated by stretch.
- afferent fibers: sensor yafferent fibers that passes with parasympathetic pelvic nerve.
- Central controller: sacral segment 2,3, and 4 in spinal cord.
- efferent fiber: parasympathetic this lead to contraction of detrusor muscle and relaxation of internal sphincter.



# Higher centres effects on micturition

- Facilitation of the micturition reflex through facilitatory areas in pons and cortex.
- Inhibition of the micturition reflex through inhibitory areas in mid brain and cortex.
- Control of external sphincter.



Flow of urine into urethra and stimulation of stretch receptors



Afferent impulses via pelvic nerve



Inhibition of pudendal nerve



Relaxation of external sphincter



Voiding of urine

Filling of urinary bladder



Stimulation of stretch receptors



Afferent impulses pass via pelvic nerve



Sacral segments of spinal cord



Efferent impulses via pelvic nerve



Contraction of detrusor muscle and relaxation of internal sphincter



Flow of urine into urethra and stimulation of stretch receptors





# Abnormalities of micturition

## Atonic bladder:

- Destruction of sensory fibers will abolish the reflex.
- the bladder will fill to capacity and overflow of drops of urine will take place, this is known as overflow incontinence.



# Abnormalities of micturition

## Automatic bladder:

Damage to the spinal cord above sacral segment will abolish the higher center effects on micturition.

During the first week micturition reflex will be suppressed due to spinal shock.

After a week micturition reflex will come back to normal and the bladder can empty periodically.

